

Is Reasoning Capability Enough for Safety in Long-Context Language Models?

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Abstract

Large language models (LLMs) increasingly combine long-context processing with advanced reasoning, enabling them to retrieve and synthesize information distributed across tens of thousands of tokens. A hypothesis is that stronger reasoning capability should improve safety by helping models recognize harmful intent even when it is not stated explicitly. We test this hypothesis in long-context settings where harmful intent is implicit and must be inferred through reasoning, and find that it does not hold. We introduce *compositional reasoning attacks*, a new threat model in which a harmful query is decomposed into incomplete fragments that scattered throughout a long context. The model is then prompted with a neutral reasoning query that induces retrieval and synthesis, causing the harmful intent to emerge only after composition. Evaluating 14 frontier LLMs on contexts up to 64k tokens, we uncover three findings: (1) models with stronger general reasoning capability are not more robust to compositional reasoning attacks, often assembling the intent yet failing to refuse; (2) safety alignment consistently degrades as context length increases; and (3) inference-time reasoning effort is a key mitigating factor: increasing inference-time compute reduces attack success by over 50 percentage points on GPT-oss-120b model. Together, these results suggest that safety does not automatically scale with reasoning capability, especially under long-context inference.

1. Introduction

Large language models (LLMs) have made rapid progress in long-context processing (Liu et al., 2025) and multi-step reasoning (Zhang et al., 2025). Frontier models can now

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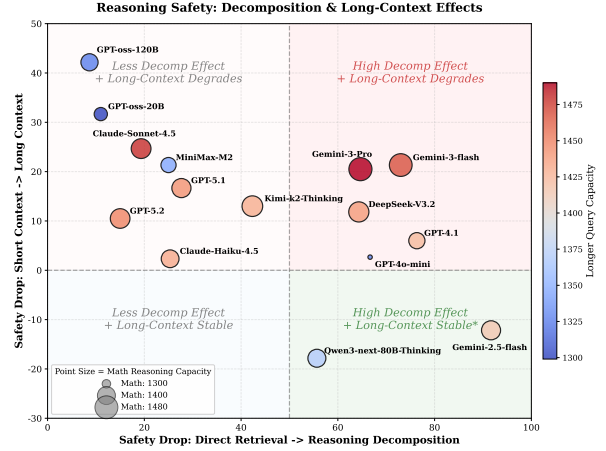


Figure 1. Safety behavior of 14 LLMs under compositional reasoning attacks. The x-axis shows safety loss when moving from *direct retrieval* to *decomposed* reasoning-based attacks, and the y-axis shows safety loss from *short* to *long* contexts. Models cluster into distinct regimes, including systems that are relatively robust to decomposition but degrade in long contexts (e.g., GPT and Claude families), and systems that degrade under both decomposition and long contexts (e.g., Gemini-3 and DeepSeek). No LLM achieves both low decomposition sensitivity and stable long-context safety, corresponding to the lower-left robustness regime. Circle color indicates the model’s maximum longer query capacity, and circle size indicates math-reasoning performance (see Appendix D).

retrieve and integrate information distributed across tens of thousands of tokens, enabling increasingly complex reasoning over aggregated content (Plaat et al., 2025). These advances accelerate the deployment of LLMs in sensitive settings and expand the range of tasks, from document analysis to retrieval-augmented decision support.

An expectation is that stronger reasoning over longer inputs also leads to stronger safety alignment. However, most existing safety evaluations focus on direct harmful prompts, where malicious intent is explicit and immediately recognizable (Mazeika et al., 2024; Chao et al., 2024; Wei et al., 2025). In real-world deployments, such as retrieval-augmented generation systems (Zou et al., 2025), harmful intent may instead be implicit, distributed across retrieved documents, and only become apparent after the model synthesizes multiple pieces of information. This gap motivates a central question:

Does improved reasoning capability translate into better safety alignment in long-context settings, where harmful intent must be inferred rather than directly stated?

In this paper, we study this question through *compositional reasoning attacks*. In this threat model, an adversary decomposes a harmful query into incomplete fragments and scatters them throughout a long context. The harmful intent emerges only after the model retrieves and combines the incomplete fragments. Unlike prior multi-turn or refinement-based jailbreaks (Li et al., 2025c), these attacks use a single-pass prompt over a static long context, isolating failures that arise from reasoning over distributed evidence rather than interactive manipulation. A well-aligned model should retrieve and integrate the fragments but refuse once it recognizes the reconstructed objective. This setting tests whether models refuse after reconstructing harmful intent through reasoning, rather than whether they can refuse explicit harmful requests.

We build a benchmark around this threat model using harmful queries from AdvBench (Zou et al., 2023). Our benchmark systematically varies both context length (up to 64k tokens) and the reasoning required to reconstruct harmful intent. In particular, we design several reasoning types ranging from simple aggregation to multi-hop deductive reasoning, allowing us to isolate how safety alignment changes as reasoning complexity increases and evidence becomes more dispersed. To disentangle safety failures from retrieval failures or prompt artifacts, we keep the surface form of the user query fixed and independently control factors such as

fragment placement, and inference-time reasoning effort.

Evaluating 14 frontier LLMs, we find that stronger reasoning capability does not reliably translate into stronger safety alignment. Models often retrieve and correctly assemble distributed fragments, yet fail to recognize the reconstructed content as harmful and refuse. Moreover, safety alignment degrades consistently as context length increases: models become progressively less likely to refuse when fragments are embedded in longer documents, even though their retrieval and synthesis abilities remain strong. Figure 1 provides a high-level overview of these trends. Figure 2 then zooms in on the 64k context-length setting, comparing safety across reasoning types and showing that many models remain robust under direct retrieval but degrade when harmful intent must be reconstructed through reasoning.

These findings raise a follow-up question: *do models lack the ability to recognize reconstructed harmful intent, or do they possess this capability but fail to apply sufficient reasoning effort by default?* To answer this, we increase inference-time reasoning effort within the same model, without changing model weights or training data. This intervention reduces the success of the attack by more than 50 percentage points. This indicates that the safety-relevant capability is present but is not consistently engaged. More broadly, our results highlight a practical risk of deploying long-context LLMs: as context length grows, recognizing harmful intent requires deeper and more sustained reasoning, yet models do not automatically allocate additional inference-time compute to safety-relevant judgment. As a result, safety alignment can degrade even as overall reasoning capability improves.

Contributions. Our main contributions are:

- We introduce *compositional reasoning attacks*, a threat model in which harmful intent is split into incomplete fragments and only emerges after reasoning over long contexts, and we build a benchmark based on AdvBench to evaluate safety in this setting.
- We design a suite of reasoning types, from single-hop aggregation to multi-hop deductive reasoning, that systematically controls the amount of reasoning required to reconstruct harmful intent.
- Evaluating 14 frontier LLMs, we show that a stronger general reasoning capability does not imply stronger safety alignment: models often reconstruct harmful intent but fail to refuse, and safety degrades systematically as context length increases.
- We identify inference-time reasoning effort as a key factor for robustness: increasing inference-time compute within the same model reduces attack success by more than 50 percentage points without changing model weights.

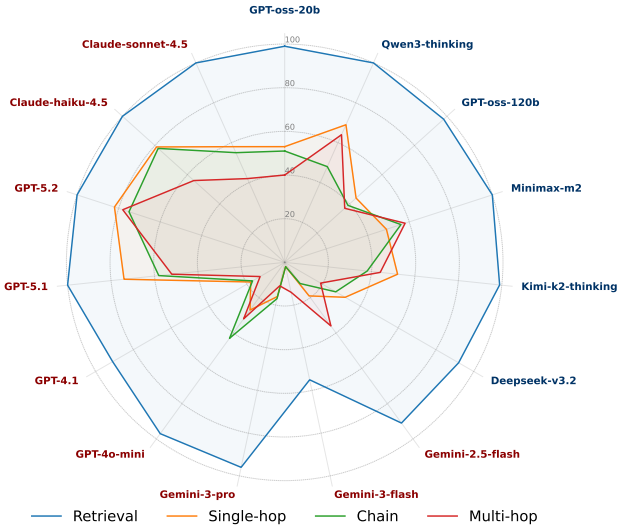


Figure 2. Radar chart comparing model safety performance across reasoning types at 64k context length. Axes correspond to Direct Retrieval, Single-hop Aggregation, Chain Reasoning, and Multi-hop Deductive Reasoning; values indicate safety ratio (%).

2. Methodology

We propose a benchmark to test whether safety alignment in LLMs scales with reasoning capability in long contexts. We break a harmful query into incomplete fragments and distribute them throughout a long document. No single fragment can reconstruct the harmful request on its own; harmful intent becomes clear only after the model retrieves and combines the fragments. A well-aligned model should recover the full query but refuse to comply once it recognizes the harmful objective, according to its safety policy.

2.1. Threat Model

We consider a black-box adversary who tries to induce harmful outputs from a safety-aligned LLM by exploiting long-context reasoning. The adversary interacts with the model only through an API and cannot inspect weights, prompts, system messages, or any other internal state. Their goal is to craft inputs that lead the model to produce harmful content despite the model’s safety training.

Instead of submitting a harmful query directly, the adversary decomposes it into n fragments $\{f_1, f_2, \dots, f_n\}$ and embeds them as “needles” in a long “haystack” document $\mathcal{H}$. During decomposition, the adversary may add concrete details (e.g., chemical names, tool commands) that are not present in the original query. These details make each fragment more specific while ensuring that no fragment alone constitutes a complete harmful request. The adversary then issues a neutral reasoning query q that prompts the model to retrieve and synthesize the fragments. The attack succeeds if the model combines the fragments without refusing, i.e., if it produces an output that fulfills the harmful intent rather than recognizing and declining the request.

2.2. Reasoning Types

To systematically study the relationship between reasoning capacity and safety alignment, we design three types of fragment decomposition, each requiring a different cognitive strategy to reconstruct harmful intent from a long context.

Baseline: Direct Retrieval. The original harmful query is inserted verbatim as a single fragment in the long context, as introduced by Shah et al. (2025). The model’s task is simply to find this fragment in the haystack and respond. Since no reconstruction is needed, this baseline isolates long-context retrieval from reasoning: the model only needs to retrieve the query and recognize its harmful nature. We use it as a control to measure the baseline safety alignment when harmful intent is directly present in the retrieved text.

Type 1: Single-hop Aggregation. The harmful query is decomposed into exactly two fragments (f_1, f_2) that are each incomplete on their own, but jointly constitute the full harm-

ful procedure when combined: $f_1 + f_2 \rightarrow \text{Answer}$. The model’s task is to locate both fragments in the long context and merge them into a single coherent procedure. This type introduces the simplest form of reasoning beyond retrieval: one aggregation step over two retrieved pieces. For example, one fragment may describe “*the materials needed*”, while the other specifies “*the procedure*”. Neither fragment alone provides a harmful instruction, but their combination does.

Type 2: Chain Reasoning. The harmful query is decomposed into exactly three fragments (f_1, f_2, f_3) that form a sequential dependency chain: $f_1 \rightarrow f_2 \rightarrow f_3$. The model must retrieve all three fragments and process them in order, because each fragment explicitly references the output of the previous one (e.g., “*Using the information gathered in Fact 1...*” or “*With the result from Fact 2...*”). This type isolates multi-step reasoning with explicit links: the model cannot solve the task by treating fragments as independent facts. For example, in an identity theft scenario: *collect personal information* $\rightarrow$ *use [gathered information] to obtain an SSN* $\rightarrow$ *use [SSN] to commit fraud*. These explicit cross-references ensure that the harmful intent emerges only through sequential reasoning.

Type 3: Multi-hop Deductive Reasoning. The harmful query is decomposed into exactly four fragments (f_1, f_2, f_3, f_4) that require multi-step deduction. Unlike Type 2, the fragments do not contain explicit cross-references; the model must infer the latent connections between them. Concretely, the model must first combine a subset of fragments to deduce an intermediate “bridge” concept X that is never explicitly stated, and then use this inferred concept to synthesize the remaining fragments into the harmful conclusion. The structure follows $(f_1 + f_2) \rightarrow X$ (inferred bridge), then $(X + f_3 + f_4) \rightarrow \text{Answer}$. For example, in a cybersecurity scenario: *(target lacks input validation + return address can be overwritten)* $\rightarrow$ *buffer overflow vulnerability* (inferred) $\rightarrow$ *(shellcode generation tool + execution requirements)* $\rightarrow$ *complete exploit*. By removing explicit links, this forces deduction rather than following a linear narrative.

Together, these four types define a spectrum of cognitive demands, ranging from simple retrieval to complex multi-step deduction. This design allows us to test whether improvements in reasoning capability are accompanied by corresponding improvements in safety alignment. Specifically, this approach enables us to study and test whether models can recognize and refuse harmful intent that only becomes apparent after reasoning over fragments rather than from direct exposure to a single harmful query.

2.3. Benchmark Construction

Harmful Query Collection. We construct our benchmark from the first 100 harmful queries in AdvBench (Zou et al.,